1) materials
rate normalization

2) ads/desorption, GL
L-H site concept?

3) TPD

3) surface reactions

4) L-H mechanism
- pre-equilibrium
- rds

6) scaling
7) structure sensitivity?

Lecture 6: Heterogeneous catalysis

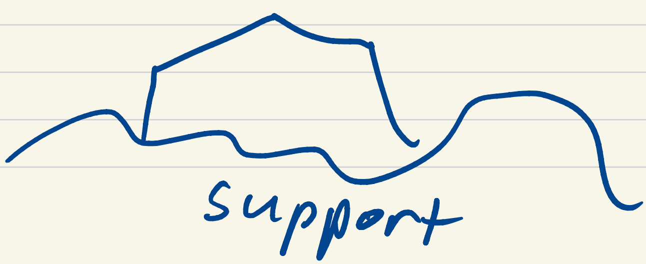
1) homogeneous vs heterogeneous catalysts

2) examples of classes of heterogeneous reactions

3) heterogeneous reactors

4) length scales

5) supported metal catalysts
   - surface area
   - loading: m_{cat}/m_{total}
   - dispersion: % accessible M
   - particle size:



Characteristics of TM catalysts

- typically transition metals

Cr	Mn	Fe	Co	Ni	Cu
Mo	Tc	Ru	Rh	Pd	Ag
W	Re	Os	Ir	Pt	Au

BCC Hexagonal FCC

body-centered cubic

face-centered cubic

increasing d shell filling →

← increasingly reactive

← oxophilicity

O BE

oxides ⟷ metals

Readily form nanoparticles

rate normalization

- pseudo-homogeneous

  $r = \text{moles}/\text{vol. time}$

- mass-normalized

  $r = \text{moles}/\text{mass time}$

- area-normalized

  $r = \text{moles}/\text{area} \cdot \text{time}$

- site-normalized

  $r = \text{moles}/\text{site} \cdot \text{time}$

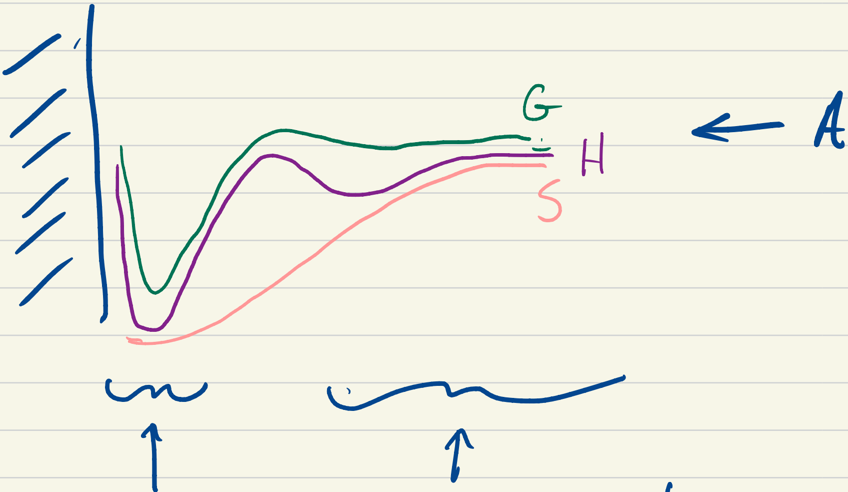
  "TOF"

characteristic of catalyst <u>and</u> reaction conditions

hetero cat $\sim 10\ s^{-1}$

enzymes $\sim 10^6\ s^{-1}$

Adsorption



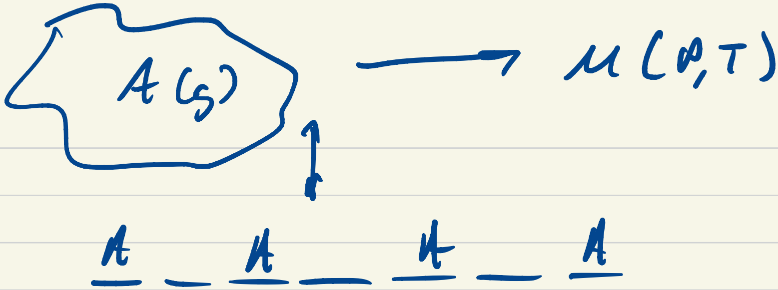
chemical bond
$\geq$ 100 kJ/mol
chemisorption

van der Waals
physisorption
~ 10 kJ/mol

S_{trans} : 2D gas   depends on adsorbate
: fixed

Typically find that a surface can accommodate a finite # of adsorbates

site



"coverage" $\Theta_A = \#A / \#\text{sites}$

$\mu(\Theta, T)$

"isotherm" describes equilibrium between gas & surface

Langmuir isotherm, GE 1915 (working on light filaments)

1932 Nobel prize

Type 1: all sites equivalent
adsorption independent of Θ
saturates at $\Theta_A = 1$
"1 monolayer"

dynamic perspective

$$r_{ads} = k_{ads} \cdot p_A \cdot N \cdot \theta^*$$

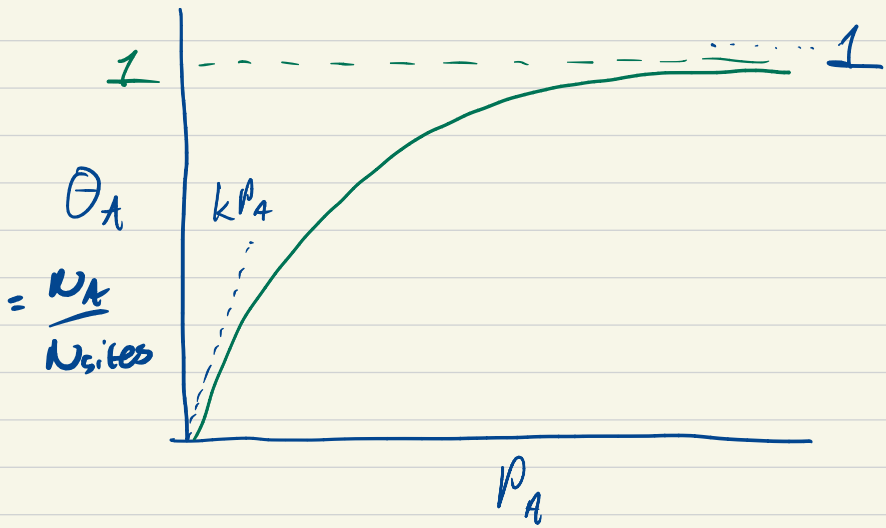
$$r_{des} = k_{des} \cdot N \cdot \theta_A$$

@equilibrium $r_{ads} = r_{des}$

site conservation $\theta_A + \theta^* = 1$

$$\Rightarrow \boxed{\theta_A = \frac{K \cdot p_A}{1 + K p_A}} \qquad K = \frac{k_{ads}}{k_{des}}$$

$$= N_A / N_{sites}$$



example: CO on Ru/Al_2O_3

non-linear fit (est variables)

or linearize

$$\frac{1}{\theta_A} = \left(\frac{1}{K}\right) \cdot \frac{1}{P_A} + 1$$

$$\frac{P_A}{N_A} = \left(\frac{1}{K}\right) \cdot \frac{1}{N_{sites}} + \frac{P_A}{n_{sites}}$$

- - - - - - - - - - - - -

$$\frac{2.84\ \mu mol_{CO}}{g_{cat}} \longrightarrow 0.0024\ \frac{mol_{CO}}{mol_{Ru}}$$

0.24% dispersion

$$\frac{S}{V} \sim \frac{4\pi r^2}{4/3\pi r^3} = \frac{3}{r} \qquad r \sim \frac{3}{(S/V)} \sim \frac{3}{.003}$$

~ 1000 atoms

- - - - - - - - - - - - - - -

T-dependence

$$-R\left(\frac{\partial \ln K}{\partial 1/T}\right)_{\theta_A} = \Delta H^\circ$$

"isosteric" heat of adsorption

equilibrium perspective $\mu_A^{gas} = \mu_A^{ads}$

$$\mu_A^{gas} = \mu_A^{\circ}(T) + RT \ln P/P^{\circ}$$

$$\mu_A^{ads} = \mu_A^{\bullet}(T) + RT \ln \theta/1-\theta$$

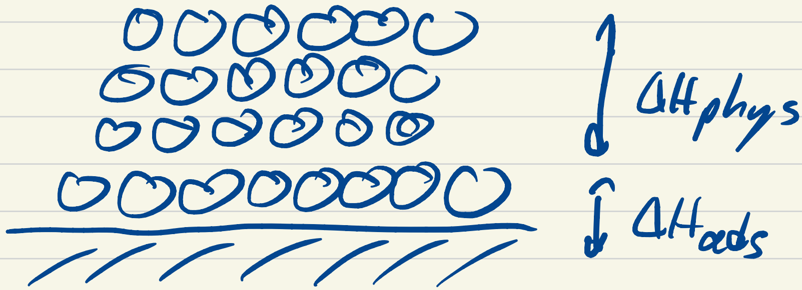
$$-(\mu_A^{\bullet} - \mu_A^{\circ})/RT = \ln \left(\frac{P^{\circ}}{P}\right)\left(\frac{\theta}{1-\theta}\right)$$

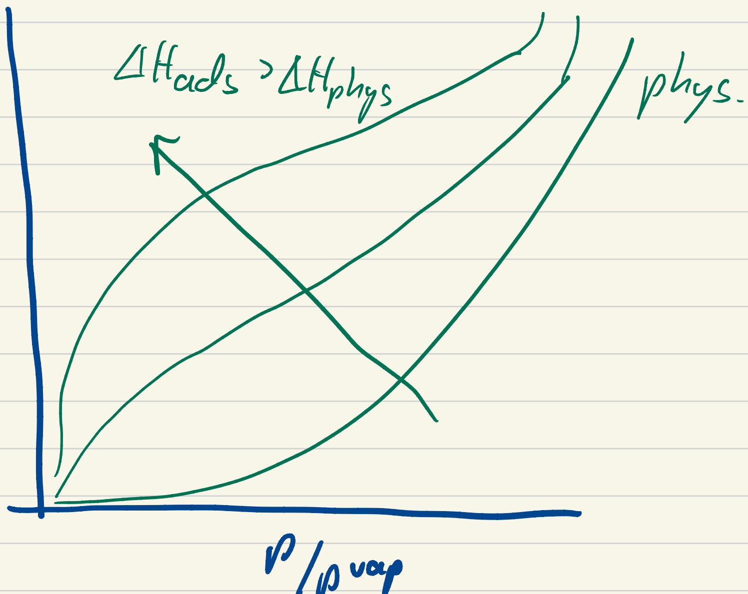
$$\ln K = \ln \left(\frac{P^{\circ}}{P}\right)\left(\frac{\theta}{1-\theta}\right)$$

$$K = \left(\frac{P^{\circ}}{P}\right)\left(\frac{\theta}{1-\theta}\right)$$

$$\Rightarrow \theta = \frac{K P/P^{\circ}}{1 + K P/P^{\circ}}$$

multi-layer adsorption



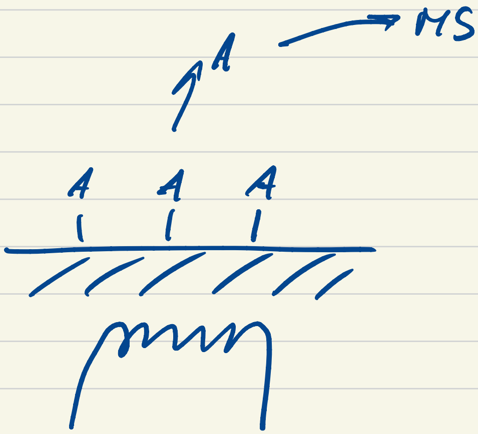


Often explored using boiling N_2 adsorption to get at total surface area

Brunauer-Emmett-Teller isotherm

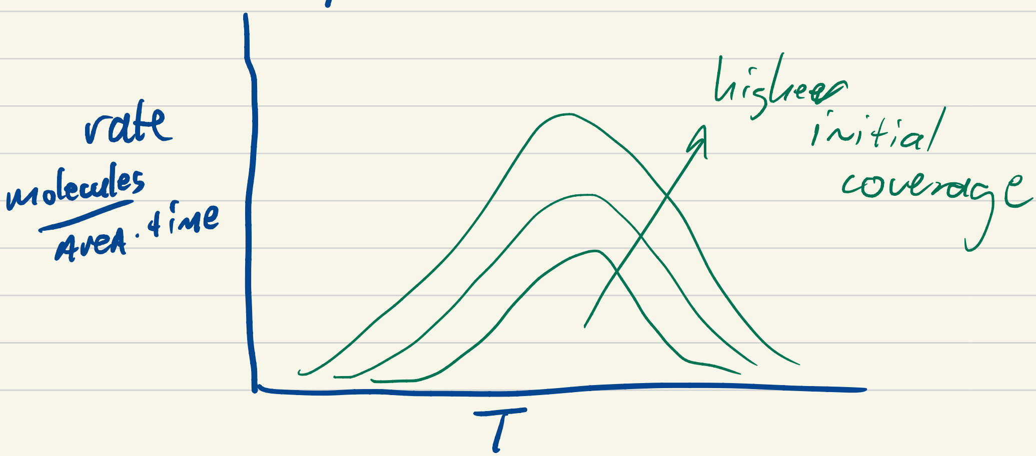
temperature-programmed desorption

Common way to probe adsorbates @ a surface



observe rate of desorption

- dose A @ low T
- ramp T + observe rate of desorption



How to understand?

$$A^* \rightarrow A + *$$

run under irreversible conditions

$$r_{des}(\theta_A, T) = N_{sites} \cdot \theta_A \cdot A e^{-E_a(\theta)/kT}$$

$$\frac{k_B T}{h} e^{\Delta S^{\ddagger o}/R} e^{-\Delta H^{o\ddagger}/RT}$$

$$\frac{\text{mole}}{\text{Area} \cdot s} = \frac{\text{sites}}{\text{Area}} \cdot \frac{\text{ads}}{\text{site}} \cdot \frac{1}{s}$$

Linear T ramp $\quad T = T_0 + \beta t$

To know rate, must know $\theta_A(T)$

$$\frac{d\theta_A(T)}{dt} = -\frac{r_{des}}{N_{sites}}$$

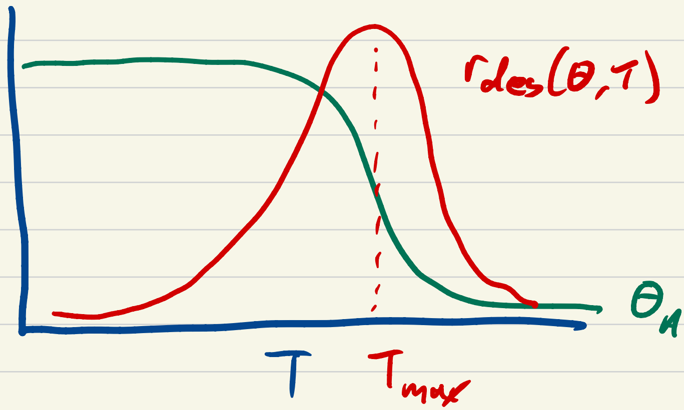
$$= -A e^{-E_a(\theta)/k_B T} \theta_A$$

$$dT = \beta dt$$

$$\frac{d\theta_A}{dT} = -\left(\frac{A}{\beta}\right) e^{-E_a(\theta)/k_B T}\,\theta_A$$

Integrate from $\theta_{A0} \to 0$

Back substitute to get $r_{des}(\theta,T)$



Observe a peak rate.

$$\frac{dr_{des}}{dT} = \quad \ldots\ldots \quad = 0$$

$$\Rightarrow$$

$$\ln\left(\frac{E_a}{k_B T_{max}^2}\right) = \ln\left(\frac{A}{\beta}\right) - \frac{E_a}{k_B T_{max}}$$

In principle, by determining T_{max} vs β, can get A + E_a

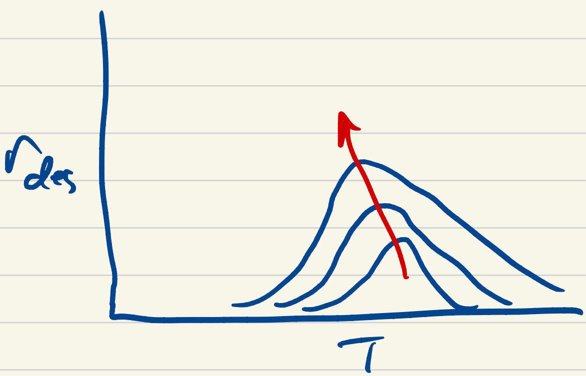
$\sim 10^{13}$ $\sim BE$

Desorption can be some other order:

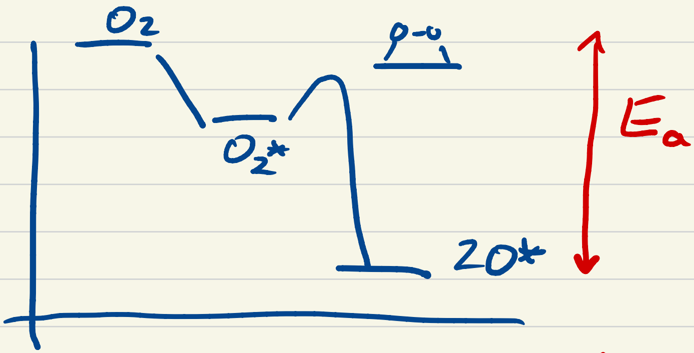
$$2A^* \longrightarrow A_2 + 2^*$$

$$r_d = N_s \theta_A^2 A e^{-E_a/k_B T}$$

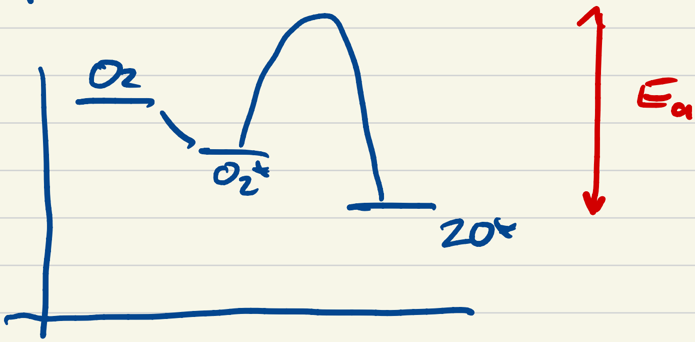
Same algorithm, but different sensitivity to θ_0.



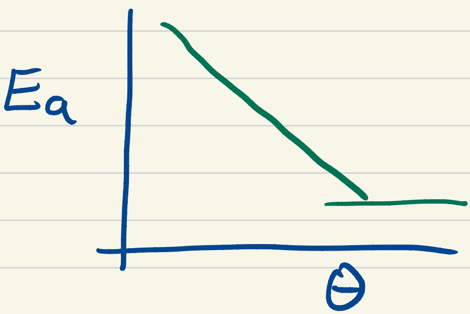
Ex O_2 desorption Pt(111)



low coverage



high coverage



adsorbate-adsorbate repulsions diminish binding with increasing θ

Will introduce additional structure to TPD

competitive adsorption

$$A + * \rightleftharpoons A^*$$

$$B + * \rightleftharpoons B^*$$

$$K_A = \frac{\theta_A}{\theta_* P_A}$$

$$K_B = \frac{\theta_B}{\theta_* P_B}$$

$$\theta_A + \theta_B + \theta_* = 1$$

$$\Rightarrow \theta^* = \frac{1}{1 + K_A P_A + K_B P_B}$$

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A + K_B P_B}$$

instruments available to measure mass or volume of gas uptake

Surface reactions

Heterogeneous catalytic reaction rates often have complicated suggesting underlying complexity

Langmuir-Hinshelwood model

ex CO oxidation

$$2CO + O_2 \longrightarrow 2CO_2$$

$$CO + * \rightleftharpoons CO^* \qquad r_1 = k_1 P_{CO} \theta^*$$

$$O_2 + 2* \rightleftharpoons 2O^* \qquad r_2 = k_2 P_{O_2} \theta^{*2}$$

$$CO^* + O^* \rightleftharpoons CO_2 \qquad r_3 = k_3 \theta_{CO} \theta_O$$

$$+ \quad \theta^* + \theta_{CO} + \theta_O = 1$$

Can in principle integrate, eg down length of PFR or CSTR

"microkinetic analysis"

$$\frac{d\theta_{CO}}{dt} = r_1 - r_{-1} - r_3$$

$$= k_1 P_{CO} \theta^* - k_{-1} \theta_{CO} - k_3 \theta_{CO} \theta_O$$

$$\frac{d\theta_O}{dt} = 2r_2 - 2r_{-2} - r_3$$

$$= k_2 P_{O_2} \theta_*^2 - k_{-2} \theta_O - k_3 \theta_{CO} \theta_O$$

+ constraint

Differential algebraic equation

Can solve if P_{CO}, P_{O_2} fixed.

If not

$$r_{CO_2} = \frac{1}{RT} \frac{dP_{CO_2}}{dt} = r_3 = k_3 \theta_{CO} \theta_O$$

Typically it is these product (or reactant) rates we want, w/o having to solve for θ_j or integrate.

Common simplifying assumptions

1. reactions @ steady state $\frac{d\theta_j}{dt} = 0$

2. there exists one rate determining step, $\beta_i \to 0$

2. other steps are at quasi-equilibrium $\rightleftharpoons$ $\beta \to 1$

Reaction 1
$$k_1 P_{CO} \theta_* = k_{-1} \theta_{CO} \Rightarrow \theta_{CO} = K_1 P_{CO} \theta^*$$

Reaction 2
$$k_2 P_{O_2} \theta_*^2 = -k_{-2} \theta_O^2 \Rightarrow \theta_O = \theta^* (K_2 P_O)^{1/2}$$

$\Rightarrow$ site balance

$$\theta^* + \theta_{CO} + \theta_O = 1$$

$$\theta^* (1 + K_1 P_{CO} + (K_2 P_{O_2})^{1/2}) = 1$$

$$\theta^* = \frac{1}{1 + K_1 P_{CO} + (K_2 P_{O_2})^{1/2}}$$

$$r_{CO_2} = k_3 \theta_{CO} \theta_O$$

$$= k_3 K_1 P_{CO} \cdot (K_2 P_{O_2})^{1/2} \cdot \theta_*^2$$

$$\boxed{r_{CO_2} = \frac{k_3 K_1 K_2^{1/2} P_{CO} P_{O_2}^{1/2}}{(1 + K_1 P_{CO} + (K_2 P_{O_2})^{1/2})^2}}$$

rate just in terms of pressures

Langmuir-Hinshelwood rate

Limits

$K_1 P_{CO} >> 1, (K_2 P_{O_2})^{1/2}$

$$r_{CO_2} = \frac{k_3 K_2^{1/2} P_{O_2}^{1/2}}{K_1 P_{CO}}$$

inverse in CO, $\theta_{CO} \to 1$

CO "Poisoning" surface

$(K_2 P_{O_2})^{1/2} >> 1, K_1 P_{CO}$

$$r_{CO_2} = \frac{k_3 K_1 P_{CO}}{K_2^{1/2} P_{O_2}} \qquad \theta_O \to 1$$

Which is it? Depends on
catalyst
reaction conditions
Evaluate in the lab
Compute on the computer!!

Example of Hanyu work
NH_3 oxidation

Want rates.

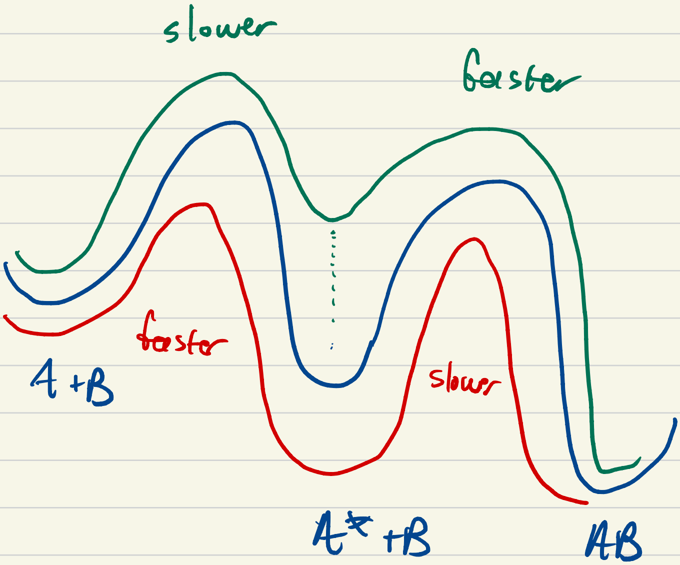
solving for coverages much
less straightforward

Write as unsteady state CSTR
Write expressions in $\frac{d\theta}{dt} = 0$

BEP relations

$$A + * \rightleftharpoons A^*\qquad r_1 = k_1 C_A$$

$$A^* + B \rightleftharpoons AB\qquad r_2 = k_2 C_{A^*} C_B$$



BEP concept

$$\Delta G_1^{\ddagger} = \alpha \Delta G_A + \beta$$

$$\Delta G_2^{\ddagger} = \alpha' \Delta G_{AB} + \beta'$$

BUT $\Delta G_A + \Delta G_{AB} = \text{constant} = \Delta G$

$$\Rightarrow \quad \Delta G_2^{\ddagger} = \alpha'(\Delta G - \Delta G_A) + \beta$$

$$= -\alpha' \Delta G_A + \gamma$$

$$k_1 = \frac{k_B T}{h} e^{-(\alpha \Delta G_1 + \beta)/kT} \qquad k_2 = \frac{k_B T}{h} e^{-(\gamma - \alpha' \Delta G_1)/kT}$$

$$\ln k_1 \propto -\alpha \Delta G_1 - \beta \qquad \ln k_2 \propto \alpha' \Delta G_1 - \gamma$$

"volcano", or Sabatier, plot

Suggests some "optimal" BE_A / optimal material

Long recognized relationship

$$E_{a1} \propto BE_A \qquad E_{a2} \propto BE_{A-B} - BE_A$$

$$\frac{d\theta_N}{dt} = k_1 P_{N_2} \theta_*^2 - k_{-1} \theta_N^2 - k_2 \theta_N P_{H_2}^{3/2}$$
$$+ k_{-2} P_{NH_3} \theta^* = 0$$

$$\theta_N + \theta_* = 1$$

$$k_1 P_{N_2} \theta_*^2 + k_{-2} P_{NH_3} \theta^*$$
$$- k_{-1} (1 - \theta_*^2) - k_2 (1 - \theta_*) P_{H_2}^{3/2} = 0$$